

JEE MAIN REVISION MASTER: COMPOUNDS CONTAINING NITROGEN

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Basicity of Amines & Hoffmann Bromamide Degradation	16 PYQs	Amine Reactions
High	Diazonium Salts & Coupling Reactions	11 PYQs	Diazonium Chemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basicity of Amines & Hoffmann Bromamide Degradation	$\text{R-CONH}_2 + \text{Br}_2 + 4\text{NaOH} \rightarrow \text{R-NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$ (Degradation by 1 Carbon)
Diazonium Salts & Coupling Reactions	$\text{ArN}_2^+ \text{Cl}^- + \text{C}_6\text{H}_5\text{SO}_2\text{Cl} \xrightarrow{\text{CuCl}_2/\text{HCl}} \text{Ar-Cl} + \text{N}_2$ (Coupling with Phenol to p-hydroxyazobenzene (Orange Dye))

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basicity of Amines & Hoffmann Bromamide Degradation (Very High Yield — 16 PYQs)

Core Principle: Hoffmann bromamide reaction converts primary amides to primary amines with ONE FEWER carbon atom: $\text{R-CONH}_2 + \text{Br}_2 + 4\text{NaOH} \rightarrow \text{R-NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$. Carbylamine test: 1° amines heated with CHCl_3 + alc. KOH produce extremely foul-smelling isocyanides (R-NC). Hinsberg's test (benzenesulphonyl chloride $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$) distinguishes 1°, 2°, 3° amines.

■ **Caution / JEE Pitfall:** Carbylamine test is given ONLY by primary (1°) amines (both aliphatic and aromatic); 2° and 3° amines do NOT respond.

■ **Speed Trick:** Basicity order in aqueous medium: For ethyl group: $2^\circ > 3^\circ > 1^\circ > \text{NH}_3$ (231); for methyl group: $2^\circ > 1^\circ > 3^\circ > \text{NH}_3$ (213).

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Basic strength of alkylamines in aqueous phase is not decided by _____." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Which of the following reaction/s does not yield an amine?..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Match the compounds given in List - I with the items given in List - II. List - I List - II (I) Benzenesulphonyl Chloride (a) Zwitterion" [Ans: N/A]

3.2 Diazonium Salts & Coupling Reactions (High Yield — 11 PYQs)

Core Principle: Diazotization: Aniline reacts with $\text{NaNO}_2 + \text{HCl}$ at $0-5^\circ\text{C}$ (273-278 K) to form benzene diazonium chloride. Sandmeyer reaction uses $\text{Cu}_2\text{X}_2 / \text{HX}$ to yield haloarenes; Gattermann reaction uses Cu / HX . Azo coupling: Diazonium salt couples with phenol in weakly alkaline medium to give orange dye (p-hydroxyazobenzene); with aniline in weakly acidic medium to give yellow dye (p-aminoazobenzene).

■ **Caution / JEE Pitfall:** Diazonium salts must be prepared and kept cold at $0-5^\circ\text{C}$; on warming with water, benzene diazonium chloride hydrolyzes immediately into PHENOL and nitrogen gas!

■ **Speed Trick:** Treatment of $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ with H_3PO_2 (hypophosphorous acid) or $\text{CH}_3\text{CH}_2\text{OH}$ reduces it cleanly to BENZENE.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Organic compound 'D' is..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Nitration of aniline in strong acidic medium gives significant amount of m-nitroaniline because..." [Ans: B]
- [2024 Shift] Q#4 (MCQ): "In the reaction, $\text{Aniline} \xrightarrow{\text{Dil.HCl}} \text{NaNO}_2 \text{P} \xrightarrow{\text{NaOH}} \text{PhenolQ}$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Organic compound 'D' is

- (A)
- (B) CH_3COOH
- (C) CH_3CHO
- (D)

Correct Answer: None

Solution Analysis: Benzene first reacts with concentrated HNO_3 and concentrated H_2SO_4 at 323-333K. This is nitration of benzene, and the product formed is nitrobenzene, marked as (A). $\text{C}_6\text{H}_5\text{NO}_2$ (A) is then reduced using Fe/HCl . The nitro group, $-\text{NO}_2$, changes into an amino group, $-\text{NH}_2$. So the product is aniline, marked as (B).

Benchmark Problem 2: KCET 2026 (Q#2)

Basic strength of alkylamines in aqueous phase is not decided by _____.

- (A) Inductive effect
- (B) Solvation effect
- (C) Steric hindrance
- (D) Hyperconjugation effect

Correct Answer: None

Solution Analysis: Basic strength of alkylamines in aqueous phase is not decided by hyperconjugation effect. So, the correct option is: Option D: Hyperconjugation effect. In aqueous solution, the basic strength of alkylamines mainly depends on: Inductive effect: Alkyl groups show +I effect and increase electron density on nitrogen. Solvation effect: The conjugate acid formed after accepting H^+ is stabilised by water molecules. Steric hindrance: Bulky alkyl groups can reduce proper solvation and affect basicity. But hyperconjugation is not an important factor for deciding the basic strength of alkylamin.